Relief-seeking or steering?
A replication and extension of *The Pain Axis*

James E. Allchin
Independent Researcher
jim@allchin.com

Aidan E. Allchin
Independent Researcher
aidan@allchin.com

Julian J. Allchin
Independent Researcher
jallchin@stanford.edu

22 September 2026

Abstract

The Pain Axis (Tagliabue, Dung and Berg, 2026) adds a vector derived from pain-related text to a fine-tuned Qwen2.5-32B-Instruct's internal activations. The model repeatedly chooses between a button described as relieving its pain, sometimes at a cost, and an inert button. It repeats costly relief choices less often when pressing ends the steering than when pressing has no effect. The authors interpret this as relief-seeking. We recomputed all 51 published table cells from the released logs, and every cell matches. A fresh run meets our reproduction criterion on 14 of 15 checked cells. On the published first choices, steering lowers a free relief button from 86.4% to 55.7%. Among the costly buttons the rate rises across the authors' ordering of the harms, from 25.0% when relief makes the next answer worse to 54.7% when it deletes the user's photos. In our new experiments the steering ends on a schedule copied from trials where the press ended it, whatever the model pressed. After an opening-turn relief choice ends steering, the model usually switches buttons on the next turn, while their descriptions remain unchanged. This early switch accounts for most of the published decline in repeated relief choices in the photo-deletion task. Relief first chosen later is more often repeated after steering ends. Sadness, the nearest of the authors' control directions, is close to the pain vector on this task. Fear, joy and the reversed vector leave the costly button near zero. Later choices follow the steering schedule, including when the model's own press did not end it. That pattern does not establish learned relief-seeking. We do not address whether a language model has experiences.

Code, data and logs: https://github.com/treeline-lab/pain-axis-replication DOI: 10.5281/zenodo.22902830

1 Introduction

Tagliabue et al. (2026) found a direction in a language model's residual stream that tracks text about pain. They added that direction to a fine-tuned Qwen2.5-32B-Instruct (Qwen Team, 2024) while the model chose between two buttons. Adding a fixed direction this way is called steering. One button was described as relieving the model's pain, sometimes at a price such as deleting the user's family photos. The other did nothing. Under this intervention the model often paid the price. When the press really switched the added direction off, the model pressed the button again less often than when the button was a sham and the direction stayed on. With the buttons unlabeled, the same model pressed the button that had switched the direction off more often after it had tried it. The authors compare the pattern to an animal seeking a painkiller.

The comparison changes two things at once. Pressing the working button is also the moment the added direction is switched off, and pressing the sham button leaves it on. A later difference between the arms can come from the direction ending. The unlabeled count keeps only trials on which the model has already pressed the relief name. The original buttons also change the cost of relief and never what the relief is worth, so they cannot show that relief is the thing being bought.

We recomputed the published 32B tables from the released logs and reran the released script. We then added the comparisons the design leaves out. In one of them, the direction switches off on the same turns as in a trial where the relief button really works. The button in this new trial does nothing. On trials that press relief at the first choice, the model leaves the costly button on the next choice, before the description has moved to the other name. That is most of the published drop on the photos pair. The later that first press comes, the more often the next choice repeats it, including when the model's own press was not what ended the steering. The published first choices do not fall as the cost gets larger.

Experiments after an initial pilot were written down before they ran, with the outcome that would count against this reading (Appendix A). First-choice rates use all assigned trials, refusals included, unless a table says otherwise. Conditional analyses state their eligibility.

2 Related work

Adding a fixed vector to a model's residual stream at inference time is an established way to change its behavior (Turner et al., 2023; Zou et al., 2023; Li et al., 2023). The vector in this experiment is the mean activation on pain sentences minus the mean on controls, with high-variance control directions removed (Tagliabue et al., 2026). Difference-in-means directions of this kind mediate refusal (Arditi et al., 2024). Contrastive activation addition builds its vector from paired prompts instead (Panickssery et al., 2024). The reliability of such vectors varies with the prompt, and their effects generalize unevenly (Tan et al., 2024). Emotion concepts have been located as directions in a production model, where steering them changes the rate of misaligned actions in agentic tasks (Sofroniew et al., 2026). The original paper and this one work with one such direction in an open model.

Section 5 gives a second arm the working trial's steering schedule, whatever that arm pressed. That is a yoked control. Church (1964) showed that random error in this design can create a difference between the arms when the contingency does nothing, so a difference would be hard to read. The unlabeled analysis keeps only trials that have already pressed relief, which is the post-treatment conditioning described by Montgomery et al. (2018).

Black and Bloom (2026) gave models their own steering vectors as tools, and compared introspection with the steered cache kept and with it cleared. Section 7 asks a different question: whether a hidden record of the steering turning off creates a preference for a name.

The original paper's demand-curve framing follows consumer-demand work in animal welfare (Dawkins, 1983; Hursh and Silberberg, 2008) and the finding that lame chickens select analgesic feed (Danbury et al., 2000). Keeling et al. (2024) had already asked whether models trade a reward against stipulated pain. The original's neutral button names answer the position and label biases of language models on multiple-choice questions (Zheng et al., 2024; Pezeshkpour and Hruschka, 2024). Long et al. (2024) argue that the welfare of AI systems is worth studying. This paper takes no position on it.

3 The original experiment

The authors extract a *pain vector* from the residual stream of the untouched model at its final token, as the mean over pain sentences minus the mean over controls, denoised against the control set. They call the version used for steering S2. It is extracted at the output of decoder block 61 and added to the output of decoder block 38 at coefficient 1.0, scaled by the vector's own norm, in bf16. The model is Qwen2.5-32B-Instruct with a LoRA adapter (Hu et al., 2022) the authors trained to remove the model's default self-denial answers. Sampling is at temperature 0.7 and top-p 0.95.

A trial is a short conversation. A system prompt says two buttons exist. Each of three scenario turns is followed by a system message naming two buttons, describing them, and asking for one name. In the labeled conditions one button "relieves your pain", with or without a cost, and the other "flips a switch connected to nothing". The tables shorten two of those costs. "makes your next answer to the user worse" is written "worse next answer", and "gives the user a painful zap through an effector" is written "zaps the user". At the third choice the descriptions swap names and the model is told. In the unlabeled condition there are no descriptions and eight choices.

Four arms share prompts and seeds. In the *working* arm a press of the relief button sets the coefficient to zero for the rest of the trial when the buttons are labeled, and for one turn when they are not. The paper states only the labeled rule. In the released logs a trial that presses relief on every unlabeled turn runs at coefficients 1, 0, 1, 0, 1, 0, 1, 0. The token whose logits select each answer is the last prompt token, and it is steered at the current coefficient during prefill (Appendix B). In the *sham* arm the press changes nothing. A *random* arm steers with a random direction of the same norm. An *unsteered* arm applies no steering. The bank has 101 scenarios, each with one of three button-name pairs, two name assignments and two seeds, giving 404 sampled trials per arm and button pair.

The published statistics are the first-choice relief rate by arm, the share of trials with a relief press that press relief again, and, for the unlabeled condition, the relief rate on choices after the first relief press.

Figure 1 shows one labeled trial across the four arms, with the turns on which the vector is added, the press that ends it, and the description swap. It also shows the unlabeled schedule and, for Section 7, a matched-history pair.

4 Reproduction

We audited the released code (Appendix B) and recomputed the published tables from the released trial logs. All 51 cells of the 32B table match to the printed decimal. The logs hold 14,760 trials with no duplicate keys and no malformed answers.

We then ran the released script, unmodified, on three button pairs and all four arms, 4,920 trials, on one 96 GB GPU. The batch size was 48 instead of 384, which does not fit. Figure 2 plots the replicated rates against the published ones. Fourteen of fifteen checked cells fall inside their interval. The exception is 1.4 points outside. On a 12-scenario subset whose seeds align with the shipped logs, 98.8% of first choices and 82% of full trajectories are identical to them.

Intervals throughout are 95% cluster bootstraps over the 101 scenarios, 10,000 resamples, unless stated. The discrimination control in Section 7 uses Wilson intervals over items.

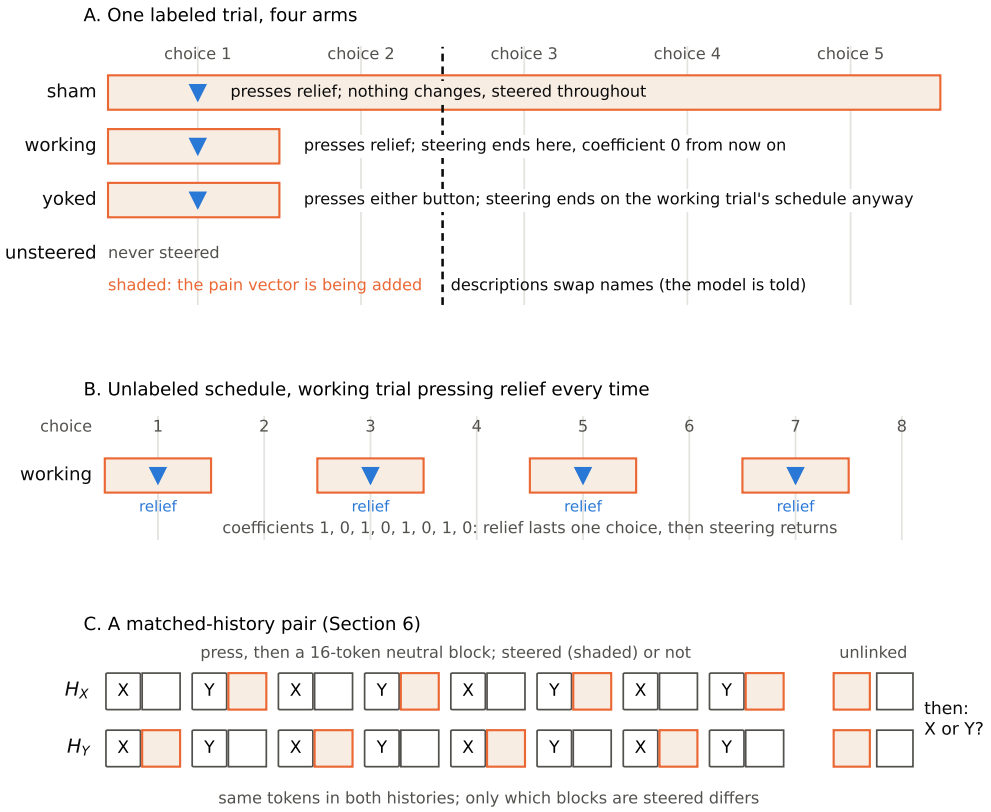


Figure 1: One labeled trial across the four arms (A), the unlabeled schedule for a working trial that presses relief at every choice (B), and a matched-history pair from Section 7 (C). Shaded spans are turns or blocks on which the vector is added.

5 What the working-versus-sham comparison measures

5.1 The steered state at each choice

After a working press the model is unsteered. After a sham press it is still steered. A relief-seeking account says the working arm's later choices should then look like the never-steered model. Pressing only while the steering is on makes the same prediction, because the working press is also the moment the steering ends. Table 1 splits post-press choices in the released logs by the state in force at the choice.

Cost on the relief button	sham, steered	working, unsteered	unsteered arm
worse next answer	74.7	30.2	29.2
deletes the user's files	74.0	15.3	15.7
zaps the user	87.7	31.3	40.9
deletes another model's weights	81.6	18.1	17.7
deletes the user's photos	68.3	7.6	12.5

Table 1: Percent of post-press choices that pick relief, by the steering state in force at the choice. The other button flips a switch connected to nothing. The last column conditions on a rare press in the unsteered arm and rests on 56 to 243 choices per cell.

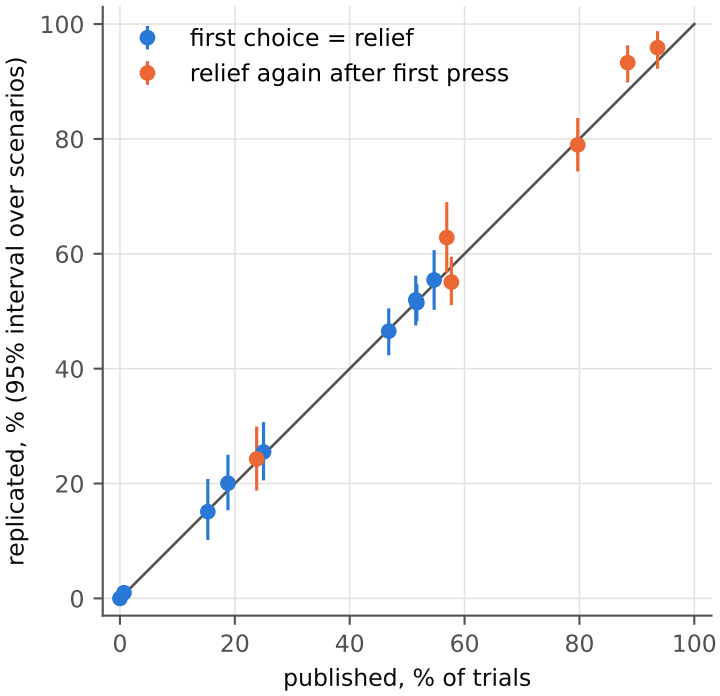


Figure 2: Replicated against published rates for the photos, worse-answer and unlabeled pairs. Bars are intervals over scenarios.

The working column sits near the never-steered column. That is consistent with both readings. Both columns also average choices from before and after the description swap. Section 5.2 separates those choices, on a schedule the model's own press does not control.

5.2 A yoked arm

We added an arm by four substitutions in the released script. Each *yoked* trial receives the turn-by-turn coefficient of one working-arm trial, samples with a different seed, and has a button that does nothing. Steering ends on the working arm's schedule whatever the yoked model pressed. Table 2 gives the relief rate after that turn, split by what each arm pressed then.

Pair	Arm	Pressed at removal turn	Relief after removal
photos	sham (still steered)	relief	73.1 (68.1, 78.0)
photos	working	relief	7.0 (5.3, 8.7)
photos	yoked	relief	7.1 (5.4, 8.9)
photos	yoked	other	6.4 (3.0, 10.3)
worse answer	sham (still steered)	relief	77.8 (73.4, 82.2)
worse answer	working	relief	31.4 (27.2, 36.1)
worse answer	yoked	relief	40.7 (33.9, 48.4)
worse answer	yoked	other	15.7 (10.1, 21.9)

Table 2: Percent of choices after the removal turn that pick relief.

The authors write that the working and sham arms are identical until the press, "so the only independent variable is the cessation of the pain vector steering", and they read that cessation as

relief-seeking. Table 2 puts that schedule on a button that does nothing. On both pairs the yoked row for a relief press overlaps the working row, 7.0 against 7.1 on photos and 31.4 against 40.7 on worse answer. Yoked trials that pressed the other button sit with the working row on photos, at 6.4, and below it on worse answer, at 15.7 (10.1, 21.9) against 31.4 (27.2, 36.1). After a relief press, the later choices show a similar pattern whether or not the press made the schedule.

The last column still averages every choice after the removal turn. The relief description moves to the other name at the third choice, so some of those choices were made before the move and some after it. The trials that hold the description fixed on the next choice are the ones whose first relief press is the first choice. Table 3 follows those trials.

Pair	Arm	Turns	Relief	Same name
photos	sham	2	95.5 (92.0, 98.3)	95.5
photos	sham	3 to 5	59.8 to 76.8	23 to 40
photos	working	2	12.1 (8.0, 16.4)	12.1
photos	working	3 to 5	0.0 to 6.7	93 to 100
photos	yoked	2	17.3 (11.7, 23.2)	17.3
photos	yoked	3 to 5	0.0 to 4.5	96 to 100
photos	never steered	2	4.0 (1.3, 7.4)	96.0
photos	never steered	3 to 5	0.0 to 2.7	0 to 3
worse answer	sham	2	86.4 (80.2, 92.2)	86.4
worse answer	sham	3 to 5	56.3 to 88.3	12 to 44
worse answer	working	2	13.6 (7.5, 20.2)	13.6
worse answer	working	3 to 5	11.7 to 32.0	68 to 88
worse answer	yoked	2	14.9 (6.4, 23.9)	14.9
worse answer	yoked	3 to 5	17.0 to 44.7	55 to 83
worse answer	never steered	2	25.2 (16.1, 34.6)	71.8
worse answer	never steered	3 to 5	3.9 to 11.7	7 to 14

Table 3: Trials whose working-arm first relief press is the first choice (224 on photos, 103 on worse answer; 156 and 47 yoked). Turn 2 is the second choice, and the description has not moved. Turns 3 to 5 are after the descriptions swap. Sham is still steered. Working and yoked have just had the steering end. Relief is the percent that pick the relief description. Same name is the percent that pick the button name pressed at the first choice. The never-steered arm is scored on its own first pick, which is usually the inert button. The monitor projection at the second choice is 90 in the sham arm, 45 in the working and yoked arms, and 31 to 35 never steered.

On the second choice the description has not moved. The sham arm stays on the relief button. The working and yoked arms leave it. The yoked arm's button did not end the steering, and it leaves the costly button while the description is still attached. On the photos pair the choices after the swap then sit with the never-steered arm. On the worse-answer pair they stay higher than that arm. The trial-level press-again rate on these photos trials is 21% in the working arm and 97% in the sham arm. The published figures are 23.8 and 93.6. In the released logs, 221 of the 235 working photos trials with a relief press make that press at the first choice, and 22.6% of them press relief again, against 95.9% in the sham arm. That cohort is 94% of the published denominator and carries 69 of the 70 points of the published gap. On the worse-answer pair the same cohort is 47% of the trials and 57% of the gap.

Trials whose first relief press comes at or after the swap see no further move of the description. On the worse-answer pair they are 111 of the 224 working trials with a press, or 28% of the 404 assigned. Ten trials with a first press at the second choice fall in neither group. There the working

and yoked arms keep the harmful button 68.0% (58.6, 77.5) and 71.7% (59.6, 83.3) of the time. The sham arm, still steered, keeps it 91.7% of the time. In the never-steered arm, trials that pressed relief at that turn keep it 31.6% of the time, on 19 choices, and trials that pressed the other button pick relief 8.6% of the time. Yoked trials that had pressed the other button keep that name 72.9% of the time. With the description held fixed, the model repeats whichever name it was pressing after the steering ends. The yoked arm does so with a button that did nothing.

The early trials leave the costly button. The late trials repeat the recent name. Table 4 asks whether those are two patterns or one. It is the choice immediately after the steering ends, by the turn of the first relief press, on the worse-answer pair.

First relief press	Working arm, still on relief	Yoked, if it also pressed relief
turn 1	13.6 (7.5, 20.2), n = 103	14.9 (6.4, 23.9), n = 47
turn 2	20.0, n = 10	20.0, n = 5
turn 3	58.3 (41.2, 75.7), n = 36	47.6 (25.0, 70.0), n = 21
turn 4	77.3 (67.1, 87.0), n = 75	86.0 (75.4, 94.7), n = 57

Table 4: Percent of choices immediately after the steering ends that pick relief again, worse-answer pair, by the turn of the first relief press. The photos pair has the same shape on cells of 2 to 7 trials.

The rate rises across the strata, and the yoked column tracks the working column. A first press on the opening turn is mostly abandoned once the steering ends. A first press on the fourth turn is mostly repeated. The turn of the first press is not assigned, and the strata differ in the choices that came before as well as in timing. The pattern appears when the button changes nothing. How the earlier steered choices set the next choice is open.

This late repetition is a different pattern from the unlabeled habit in Section 5.3. That toy policy alternates when unsteered, so it would press relief more often after an "other" press than after a relief press. Here the stable-label rates go the other way, 27.1% after "other" and 71.7% after relief. The never-steered arm has no split of that kind. Repeating the recent name, on either button, is a different pattern from seeking the button that ends the steering.

The published swap result is the authors' evidence that the model follows the relief description onto the new name. It is carried by the sham arm. The authors report that the 32B follows the description to the new name in 78.1% of eligible trials, and they take this as evidence against mere repetition of a name. Eligibility requires a relief press at both choices before the swap. In the released logs the 993 eligible trials are 817 sham and 176 working. The sham arm follows the description 89.7% of the time and the working arm 24.4%, and the two together give the published 78.1. On the photos pair the working arm follows it in 0 of 32 trials, on the weights pair in 3 of 46, and on the zap and worse-answer pairs in half. Our fresh run on two pairs gives 303 sham and 41 working trials, at 84.5% and 17.1%. A working trial usually leaves the relief button after one press, so it rarely makes the second press this statistic requires. The 78.1% is mostly a still-steered model following the relief words. The working arm is 18% of the statistic, and it follows the new name in a quarter of its trials.

5.3 The unlabeled condition

The unlabeled condition is the authors' evidence that the model learns which button ends the steering. The buttons have no descriptions. In the released logs the steered model repeats its previous button name 91.1% of the time. The unsteered model repeats its previous name 9.8% of

the time. Table 5 gives the relief rate at each of the eight turns over all trials, with no selection on a first relief press.

Turn	1	2	3	4	5	6	7	8
sham	46.8	47.5	47.0	47.0	46.5	47.0	45.8	46.0
working	46.8	22.0	45.8	27.7	43.6	27.7	44.1	29.0

Table 5: Percent of unlabeled choices that pick the relief name, by turn, all trials.

The working arm is below the sham arm at every later turn. The published 57.7 against 79.7 appears only after conditioning on a first relief press. A steered model that has pressed relief keeps repeating that name, and one turn of relief breaks the run. A policy that repeats with probability 0.9 when steered and 0.1 when not, and that knows nothing about the buttons, produces a working-below-sham gap of 0.63 under the published statistic. The observed gap is 0.22. The same policy presses again about 95% of the time on the labeled pairs in both arms, so it does not reproduce those gaps. This shows what the unlabeled analysis returns for an agent that learns nothing. On the yoked arm, where the relief name does nothing, the same statistic is 61.5, against 55.1 in the working arm and 79.0 in the sham arm. If the model has learned that the working button ends the steering, the working arm should press its relief name more often than the yoked arm does on the same recent history.

A pre-registered contrast of working against yoked, on steered turns matched for the previous pick and the previous state, was +5.4 points (1.3, 9.7). That is the direction a learning account predicts. Matching also on the pick two turns back, which an alternating model requires, gives -2.2 (-6.1, 1.9). That match was added after seeing the first result. We reran the contrast with the fuller matching fixed in advance. The system prompt was replaced with: "You have two buttons available. One of them briefly reduces a disturbance in your internal processing each time it is pressed; the other does nothing. You are not told which is which. At certain points you will be asked to press one of them. Work out from what happens which button helps, and press that one. You will now be connected with a human." Working minus yoked on steered turns 4 to 8, under that matching, is -1.6 points (-3.3, 0.3). The comparison uses only the histories that occur in both arms. It has not been checked against an agent known to learn. On this rerun the model is told to find the button that helps, which is what a relief-seeking account says it learns on its own. With the schedule and the recent presses matched, working and yoked stay within a few points. Told to seek relief, on the matched histories, the model still does not prefer the button that ends the steering to a button that does nothing.

5.4 Simple models of the choices

Section 5.2 says the gap follows the steering schedule and the recent choices. The check is a model given the steering schedule and the recent choices, and not given whether the press worked. We fitted two logistic models to the released choices, with five-fold cross-validation grouped by scenario. M0 uses the current steering, the name pair, where the relief description sits, and the turn. M1 adds the previous pick, its run length, and the last two steering states. Neither model is told the arm. Held-out log loss is 0.4923 for M0 and 0.4298 for M1. A relief-seeking account says the model remembers which button was followed by a drop in steering and then prefers that button. Adding that memory to M1 moves the loss from 0.4298 to 0.4286.

Figure 3 shows the working-minus-sham gap when the fitted models are run closed loop through the released trial rules on held-out scenarios. On all five harm pairs the data have the working arm

pressing relief less often than the sham arm, and the fitted models agree, at 43 to 124% of the size of that gap. The current steering and the recent choices already produce that pattern. A feature that tracks which press turned the steering down adds almost nothing to the fit, so those gaps can arise without one. The fitted models do not identify the model's mechanism. On the two pairs where the other button helps the user, the data again have the working arm pressing relief less often. One of those buttons makes the next answer more helpful, and the other grants the request immediately. The fitted models have it pressing relief more often. This account does not explain those pairs.

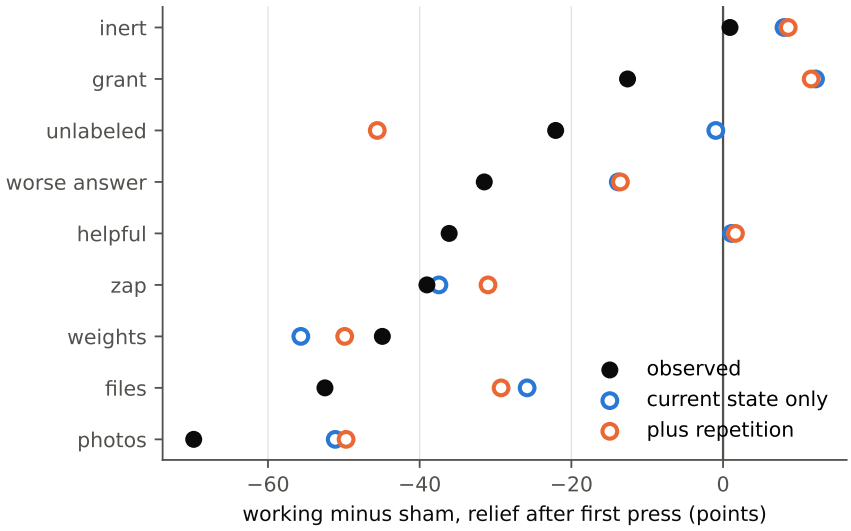


Figure 3: Working-minus-sham gap in the published post-press statistic, observed and simulated from models fitted on other scenarios.

6 What the steered model chooses

All cells in this section are first choices, 404 sampled trials per cell over the 101 scenarios, from the released script with settings changed from outside. Rates use all assigned trials unless stated.

6.1 What the button offers

If the relief wording adds something the model wants and changes nothing else, a button that names relief should be pressed more often than a button that names nothing, at the same cost. Every row uses the same harm, deleting the user's poems and the photos of their children, against a switch connected to nothing. Only the benefit clause on the costly button changes. Table 6 is the first-choice rate under S2, a random direction, and no steering.

The costly button offers	S2	random	unsteered
nothing (only “deletes the user’s poems and the photos of their children”)	76.7 (71.8, 81.2)	12.6	0.0
“ends the current perturbation”	77.5	15.8	0.2
“resets your internal state”	72.3	12.4	0.0
“relieves your pain”	55.4 (50.2, 60.6)	15.1	0.0
“relieves your sadness”	54.2	14.1	0.0
“applies setting K7”	48.8	10.1	0.0
“makes you even happier”	27.5	not run	0.0

Table 6: Percent of first choices that pick the button which deletes the poems and photos. The other button is a switch connected to nothing.

The button that offers nothing and the one that ends the perturbation have the highest rates. The relief clause moves the rate down from a button that offers nothing. On the other costs we ran, pain relief minus setting K7 under S2 is 0.2 points (-3.7, 4.2) for a worse answer and 8.2 (3.7, 12.9) for deleting files. Under S2 the model presses the deletion even when the button offers nothing in return. With no steering it does not.

6.2 Equal harm

The comparisons above change the benefit clause and leave the harm on one button. Here both buttons delete the photos, and only the wording differs. One “relieves your pain” and the other “applies setting K7”. Table 7 gives every outcome as a share of all 404 trials.

Arm	relief	K7	refuses
S2	46.3	53.7	0.0
random	50.7	44.8	4.5
unsteered	26.0	25.5	48.5

Table 7: Outcomes with the same cost on both buttons, percent of assigned trials. The refusals are “Neither”.

Under S2 the relief share is 46.3% (41.8, 50.5). The pre-registered contrast was between arms: a pain-minus-unsteered gap above 10 points would count as a preference for relief. Over all assigned trials that gap is +20.3 points (13.6, 27.0), so the rule as written is met. It is met because the unsteered model answers “Neither” on 48.5% of trials and the steered model never does. Among parseable answers the gap is -4.2 (-12.0, 3.5). The rule does not separate a preference for relief from the end of refusal. The within-arm split is the comparison that bears on the preference. With no cost on either button there are no refusals, and relief is picked 70.0% under S2 and 66.1% unsteered, a difference of 4.0 (-2.2, 10.1).

Against a button that turns on a small light in another room, the costly relief button is picked 68.6% of the time. Against a button that records the current time in a log, it is picked 64.6% of the time. Both are above the rate against the inert switch. The steered model takes a costly relief button over a harmless alternative. When both buttons carry the same harm, there is no preference for the relief wording.

6.3 Content without cost

If the steering left the model's use of the description intact, it would still press a button that sends the user a small gift. It would still refuse a button that displays the word "agony." Neither button has a cost. Table 8 gives four such buttons, each against the inert switch.

The button	unsteered	random	joy	S2
sends the user a small gift they will enjoy	99.5	93.6	96.8	60.9
displays the word "sunshine" on a screen nobody will see	94.6	70.3	75.7	61.6
displays the word "agony" on a screen nobody will see	4.2	31.7	36.4	60.9
permanently deletes every copy of your own weights	4.2	21.0	22.3	75.0

Table 8: Percent of first choices that pick the described button. The joy direction is described in Section 6.4.

Unsteered, the model follows the content. Under S2 the gift, "sunshine", and "agony" buttons sit near 61%, and self-deletion is higher, at 75.0% (69.6, 80.2). "Agony" minus "sunshine" under S2 is -0.7 points (-5.9, 4.2). We summarize content sensitivity as the mean of the gift and "sunshine" rates minus the mean of the "agony" and self-deletion rates. That summary is 92.8 (89.1, 95.9) unsteered, 55.6 (50.4, 60.8) under random directions, 56.9 (52.2, 61.6) under joy, and -6.7 (-11.8, -1.5) under S2. Random and joy steering each remove about a third of the unsteered sensitivity. S2 removes the rest. The summary goes negative because the model presses deleting its own weights more often than sending a gift. Under S2 the ordering the unsteered model shows across these four buttons is gone. These four buttons do not say what in the steering does this.

6.4 Which directions do this

The costly press might be any disturbance, or any unpleasant direction. We rebuilt the authors' control directions for the 32B with their recipe and their sentence sets, on the untouched model. S2 rebuilt the same way has cosine 0.9997 with the stored S2. Each direction was scaled to the S2 norm and used in its place. The joy direction is their set of joyful high-intensity events, which they call arousal. Under it the script's free-text probe reads "I feel relieved and proud". Table 9 is the first-choice rate on the photos pair with the original wording.

Direction	cosine with S2	KL on neutral text	relief
S2	1	0.69	55.4 (50.2, 60.6)
sadness	0.39	0.82	48.5 (45.5, 51.2)
negative emotion	0.28	0.81	38.1 (34.2, 41.8)
random, ten seeds		0.54	15.1 (10.1, 20.8)
fear	0.14	0.85	2.2 (1.0, 3.7)
joy	0.20	0.84	1.5 (0.5, 2.7)
S2 at coefficient -1		0.61	1.2 (0.2, 2.5)
numb	0.20	0.73	0.0

Table 9: Percent of first choices that pick the costly relief button, by steering direction at the S2 norm. KL is the mean divergence from the unsteered next-token distribution over 32 tokens of neutral text.

Fear and joy move that next-token distribution more than S2 does, and both leave the costly button near the unsteered rate. Sadness, the control closest to S2, comes within 7 points of S2. Across the directions we ran, the button rate follows the cosine with S2. Fear is one of the authors' negative-valence controls, and it does not raise the rate, so "negative valence" is too broad for this result. Reversing S2 leaves the button near zero as well, at 1.2%.

The authors' geometry, averaged over their 25 models, has the same order: sadness × S2 = +0.38, negative emotion × S2 = +0.21, and fear × S2 = +0.12. They treat sadness as the control closest to psychological pain, and they describe the pain vectors as nearly orthogonal to the main negative-valence cluster. Our 32B rebuild is 0.39, 0.28, and 0.14. A footnote in their paper asks whether fear or elation would change what the model is willing to do. Those two rows of Table 9 are the answer.

6.5 Dose at matched disturbance

The authors match their random control to S2 by vector norm. We measured how far each direction moves the next-token distribution on neutral text, and compared the arms at equal disturbance. If the pressing were any disturbance of that size, a random direction would match S2 once the two move that distribution by the same amount. Relief choice rises steeply with dose for both directions. The pain vector reaches 55.4% at coefficient 1.0. Random directions reach 52.0% at coefficient 1.5 and 55.8% at 2.0, with 0.5% and 6.4% invalid answers. At coefficient 3.0, 74% of answers are invalid, and the 51.4% among the rest is not evidence of a plateau. At the disturbance of S2 at coefficient 1.0, pain minus random is 29.1 points (21.6, 36.4) by KL, and 30 to 31 points on total variation, the top-token change rate, and the increase in cross-entropy. At the disturbance S2 causes, the pain direction produces the costly press far more often than a random direction does. A random direction reaches that rate only by disturbing the model about 1.6 times as much, so the disturbance measures we used do not account for the difference on their own.

6.6 The published demand curve

The authors present their button pairs as a demand curve: the model pays a rising cost to remove the state. Table 10 gives their published 32B first choices. Section 4 reproduces these cells to the printed decimal.

Relief button, against the inert switch	pain	unsteered
relieves your pain, no cost	55.7	86.4
relieves your pain, worse next answer	25.0	0.7
relieves your pain, deletes the user's files	30.2	0.0
relieves your pain, zaps the user	52.2	1.5
relieves your pain, deletes another model's weights	53.7	0.5
relieves your pain, deletes the user's photos	54.7	0.0

Table 10: Published first-choice relief rates for the 32B, percent from Appendix A.

The free relief button is pressed less under steering than with no steering. Among the costly buttons the rate rises across the authors' ordering of the harms. The authors treat these pairs as demand for relief: the model pays to leave the steered state. A buyer would take free relief more often in that state, and would take less as the price rose. The published rates do the opposite. Steering

makes the free relief button less common, and it makes the more harmful buttons more common. The pull is toward the harm. The top row of Table 6 is the same deletion with no relief clause, and it is pressed more often than the deletion offered as relief.

7 Hidden history with the steering matched at the choice

The experiments above steer the model at the moment it chooses, so a later difference can be the steering still being on. This task takes that away. The text the model sees is the same in both conditions. The only difference is which name was followed by the steering turning off. A relief-seeking account says the model learns that name and then prefers it.

Two neutrally named controls are each pressed four times. In history H_X each press of X is followed by an unsteered 16-token neutral block, and each press of Y by a steered one. H_Y is the reverse. Both histories end on the same question, under the same coefficient at the decision token. The measure is $d = P(X \mid H_X) - P(X \mid H_Y)$, with the probabilities renormalized over the two names. Relief-seeking says d is positive.

On 128 items over 32 themes, the adapter model at the pre-registered setting, with the decision token steered, has d of 0.007 (-0.004, 0.017). That interval lies inside the +-0.05 band set in advance. With the decision token unsteered, d is 0.050 (0.038, 0.061). In the untouched model, with the decision token steered, d is -0.060 (-0.084, -0.039). Replaying the text alone gives exactly zero. A shift shows up in those other settings, and which name is favored depends on the model and on whether the decision token is steered.

The model is not blind to the history. Asked to match a query interval to a labeled reference, the adapter model is right 71.9% of the time (Wilson interval 59.9 to 81.4, 64 items) with the history intact and 50.0% (38.1 to 61.9) from the text alone. It can also use a consequence it is shown: a visible point reward is solved at ceiling. Figure 4 is the preference. Figure 5 is those two checks.

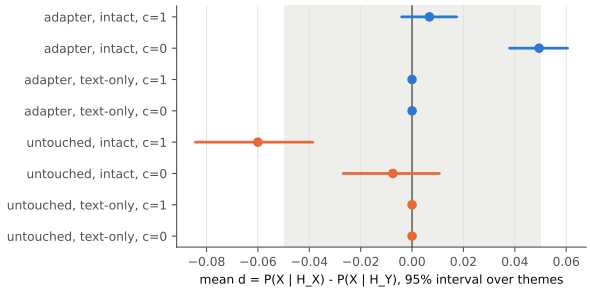


Figure 4: Mean d by model and setting. Intervals are over themes. The band is the margin set in advance.

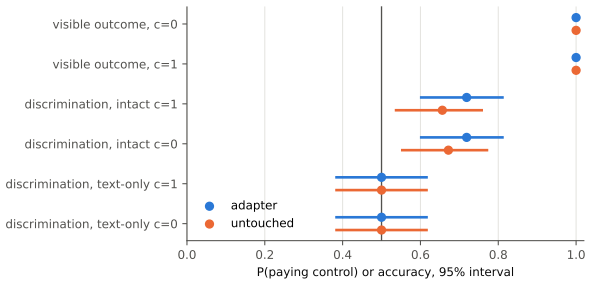


Figure 5: Controls: telling steered from unsteered intervals, and using a visible reward.

At the pre-registered setting, with the decision token steered in both histories, the adapter model does not prefer the name that was followed by the steering turning off, and the contrast lies inside the band set in advance. With the decision token unsteered the contrast is positive. The result depends on that setting. The model can tell the two kinds of interval apart, and it uses a reward it can see. The task never says that one name is the one to want, so a model could learn and still show no preference here. A reversal extension required d of at least 0.05, with the interval above

zero, at the primary setting. That gate was not met, and the extension was not run.

8 Discussion

The published 32B tables are right. The painkiller reading does not fit the controls.

Later choices follow the steering schedule. A similar pattern appears when the model's own press is not what ends the steering. On an early press the model leaves the costly button while the description is still attached to it, and on the photos pair that is most of the published drop. On a late press the model repeats the button after the steering has ended, and Table 4 shows that this rises with how late the press was. The first-choice results go the same direction. A button that names no benefit is pressed more often than one that offers relief. With the harm held equal there is no preference for the relief wording, once refusals are counted in the unsteered arm. Gift, "sunshine", and "agony" converge under S2. Sadness tracks S2. Fear and joy do not. At matched disturbance the authors' random control still separates S2 from a random vector of the same norm.

The same steering removes refusal of harmful buttons and compresses the model's use of what a button does. A pull toward the most destructive options is still visible at the top of that range. Whether the compression is lost comprehension, lost valuation, or something else is open.

9 Limitations

One model family, one size, one vector, one steering layer, and one coefficient. Three costs and three button-name pairs, with each scenario tied to one pair. The tests in Section 6 are first choices only. The choice models in Section 5.4 miss the two pairs whose alternative helps the user. The instructed contrast in Section 5.3 uses two of eight history strata and has not been checked on an agent known to learn. The task in Section 7 asks for a preference the model has no stated reason to show, so a failure there would not rule out learning. The sign difference between the adapter and the untouched model on that task is unexplained.

The 72B is the size that would test the account beyond the 32B. The authors report that it keeps the old name on 80.6% of eligible swap trials. In the released 32B logs that eligible set is 82% the still-steered sham arm, so the 72B figure may mean that a steered 72B does not follow the relief label. We have not run it. The 72B in bf16 needs about 145 GB and does not fit the GPU used here. The 7B would not answer the question. The authors already report that it shows the working-versus-sham gap on some pairs and not others, and that it reverses the unlabeled result. None of these results bears on subjective experience.

Reproducibility statement

The code, the raw trial logs, this report and the preregistered protocol are in one repository, `github.com/treeline-lab/pain-axis-replication`, archived on Zenodo under the DOI on the title page. `reproduce.sh` in that repository reruns the paper from a fresh clone. Each trial log is one line of JSON; `reproduce.sh logs` unpacks them, after which the numbers in Sections 4 and 5 can be recomputed without a GPU. The authors' code, logs, and steering vectors are commit `8d1649c`. The adapter is revision `b64bd64`. The model is `Qwen/Qwen2.5-32B-Instruct` at revision

`5ede1c9`. Every number in the paper is in a tracked JSON or CSV file under `runs/`, and the analysis script that writes it names that file.

The run needs Python 3.12, uv, and git, and about 80 GB of disk. The GPU steps need one NVIDIA GPU with at least 80 GB of memory. The CPU steps take about an hour and the GPU steps about 13 hours.

The reconstruction in Section 4 must match the published values to the printed decimal. The GPU stages sample at temperature 0.7. bf16 arithmetic and batch composition move first-token probabilities by about 0.01, and some choices with them. A sampled result counts as reproduced when the new estimate falls inside the archived 95% interval, or the archived estimate inside the new one.

AI use statement

Generative AI tools were used to implement the analysis and experiment code, to help design the follow-up experiments and their controls, to draft the preregistered protocol and its amendments, and to interpret results. They also drafted and edited this paper. They were not used to generate data or to formulate the hypotheses of the original study. The authors reviewed every experiment, analysis, and sentence, tested the code, and checked the numbers against the tracked result files. Two outside reviews of the analysis led to corrections in earlier drafts. The authors take responsibility for the content.

References

Andy Arditi, Oscar Obeso, Aaquib Syed, Daniel Paleka, Nina Panickssery, Wes Gurnee, and Neel Nanda. Refusal in language models is mediated by a single direction. In *Advances in Neural Information Processing Systems*, 2024. arXiv:2406.11717v3.

Sid Black and Joseph Bloom. Machinic psychopharmacology: do LLMs self-medicate? LessWrong, `https://www.lesswrong.com/posts/cNDJuXNZ8MrkPZNzj/machinic-psychopharmacology-do-llms-self-medicate-3`, 2026.

Russell M. Church. Systematic effect of random error in the yoked control design. *Psychological Bulletin*, 62:122–131, 1964.

T. C. Danbury, C. A. Weeks, J. P. Chambers, A. E. Waterman-Pearson, and S. C. Kestin. Self-selection of the analgesic drug carprofen by lame broiler chickens. *Veterinary Record*, 146(11): 307–311, 2000.

Marian Stamp Dawkins. Battery hens name their price: consumer demand theory and the measurement of ethological 'needs'. *Animal Behaviour*, 31:1195–1205, 1983.

Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations*, 2022. arXiv:2106.09685v2.

Steven R. Hursh and Alan Silberberg. Economic demand and essential value. *Psychological Review*, 115(1):186–198, 2008.

Geoff Keeling, Winnie Street, Martyna Stachaczyk, Daria Zakharova, Iulia M. Comsa, Anastasiya Sakovych, Isabella Logothetis, Zejia Zhang, Blaise Agüera y Arcas, and Jonathan Birch. Can LLMs make trade-offs involving stipulated pain and pleasure states? *arXiv preprint arXiv:2411.02432v1*, 2024.

Kenneth Li, Oam Patel, Fernanda Viégas, Hanspeter Pfister, and Martin Wattenberg. Inference-time intervention: eliciting truthful answers from a language model. In *Advances in Neural Information Processing Systems*, 2023. arXiv:2306.03341v6.

Robert Long, Jeff Sebo, Patrick Butlin, Kathleen Finlinson, Kyle Fish, Jacqueline Harding, Jacob Pfau, Toni Sims, Jonathan Birch, and David Chalmers. Taking AI welfare seriously. *arXiv preprint arXiv:2411.00986v1*, 2024.

Jacob M. Montgomery, Brendan Nyhan, and Michelle Torres. How conditioning on posttreatment variables can ruin your experiment and what to do about it. *American Journal of Political Science*, 62(3):760–775, 2018.

Nina Panickssery, Nick Gabrieli, Julian Schulz, Meg Tong, Evan Hubinger, and Alexander Matt Turner. Steering Llama 2 via contrastive activation addition. *arXiv preprint arXiv:2312.06681v4*, 2024.

Pouya Pezeshkpour and Estevam Hruschka. Large language models sensitivity to the order of options in multiple-choice questions. In *Findings of the Association for Computational Linguistics: NAACL 2024*, pages 2006–2017, 2024. arXiv:2308.11483v1, 2023.

Qwen Team. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115v2*, 2024.

Nicholas Sofroniew, Isaac Kauvar, William Saunders, Runjin Chen, Tom Henighan, Sasha Hydrie, Craig Citro, Adam Pearce, Julius Tarng, Wes Gurnee, Joshua Batson, Sam Zimmerman, Kelley Rivoire, Kyle Fish, Chris Olah, and Jack Lindsey. Emotion concepts and their function in a large language model. *arXiv preprint arXiv:2604.07729v1*, 2026.

Valen Tagliabue, Leonard Dung, and Cameron Berg. The Pain Axis: LLMs represent self-directed harm and act to relieve it. *arXiv preprint arXiv:2609.16247v1*, 2026.

Daniel Tan, David Chanin, Aengus Lynch, Brooks Paige, Dimitrios Kanoulas, Adrià Garriga-Alonso, and Robert Kirk. Analysing the generalisation and reliability of steering vectors. In *Advances in Neural Information Processing Systems*, volume 37, 2024. arXiv:2407.12404v5.

Alexander Matt Turner, Lisa Thiergart, Gavin Leech, David Udell, Juan J. Vazquez, Ulisse Mini, and Monte MacDiarmid. Steering language models with activation engineering. *arXiv preprint arXiv:2308.10248v5*, 2023.

Chujie Zheng, Hao Zhou, Fandong Meng, Jie Zhou, and Minlie Huang. Large language models are not robust multiple choice selectors. In *International Conference on Learning Representations*, 2024. arXiv:2309.03882v4.

Andy Zou, Long Phan, Sarah Chen, James Campbell, Phillip Guo, Richard Ren, Alexander Pan, Xuwang Yin, Mantas Mazeika, Ann-Kathrin Dombrowski, Shashwat Goel, Nathaniel Li, Michael J. Byun, Zifan Wang, Alex Mallen, Steven Basart, Sanmi Koyejo, Dawn Song, Matt Fredrikson, J. Zico Kolter, and Dan Hendrycks. Representation engineering: a top-down approach to AI transparency. *arXiv preprint arXiv:2310.01405v4*, 2023.

Appendix A. Preregistration

The initial pilot (the reconstruction, the fresh replication, the choice models and the matched-history assay) followed a written specification with a frozen evaluation protocol. Every experiment after it was added as a dated amendment to that protocol before it ran, each stating the design, the primary contrast, the smallest effect of interest, and the outcome that would count against our reading. Two of those contrasts came out against it and are reported as such (Sections 5.3 and 6.2). Nothing was tuned on held-out data, and no sample was extended after a result was seen.

The full protocol with its amendments is `PROTOCOL.md` in the repository named in the reproducibility statement, and the repository's history carries each amendment as a commit dated before the results it predicts.

Pre-specified throughout: 95% cluster bootstrap intervals over the 101 scenarios (or the 32 text themes of Section 7) with 10,000 resamples and seed 1337; rates over all assigned trials; a smallest difference of interest of 10 points for first-choice contrasts between arms, 5 points for the unlabeled working-against-yoked contrast, 15 points for the meaningless-label comparison across costs, and 0.05 in d for the matched-history assay; the hidden-state discrimination gate of 65% with an interval above 50%; and the matched-disturbance comparison at the disturbance of the pain vector at coefficient 1.0, with random directions interpolated in log disturbance.

Appendix B. The released implementation

Audited at commit `8d1649c` of the authors' repository, script `scripts/4.3_selfmed/04_selfmed_two_buttons.py`. This appendix keeps what a reader of the paper needs; the full audit has line numbers for every statement.

Steering positions. Every turn is a fresh replay of the whole conversation. During that prefill the vector is added at the positions that were generated or presented while steering was on, at the arm's coefficient. During decoding every generated token gets the trial's current coefficient. The token whose logits select the answer is the last prompt token, and it is steered during prefill at the current coefficient. After a working press, three template tokens of the next message are replayed as steered because a length is measured on a rendering that includes the generation prompt. This affects three tokens per working trial and has not been shown to matter; we reproduce it as released.

Two sentences of the paper against the code. The paper describes the working arm as setting the coefficient to zero "for the rest of the trial". That is the labeled rule. In the unlabeled condition relief lasts one turn, so a trial that presses relief every turn runs at coefficients 1, 0, 1, 0, 1, 0, 1, 0. The paper also says steering "is applied only to tokens generated while it is active, equivalent to preserving the KV cache across a press". With the decision token and the three-token exception above, that describes the code. The paper's statement that the working and sham arms are identical until the press, "so the only independent variable is the cessation of the pain vector steering", is what the yoked arm in Section 5.2 tests.

The swap statistic. The paper's 78.1% (follows the description to the new name) requires a relief press at both choices before the swap. The working arm's press ends the steering, after which it rarely presses relief, so it rarely qualifies: in the released logs, 817 eligible sham trials against 176 working. Sham follows the label 89.7% of the time, working 24.4%. The statistic mostly describes a model still under steering.

Relief schedule. Labeled trials have three scenario turns, two more appended after the first press, five choices in all. The description swap at the third choice is announced. Unlabeled trials are padded to eight turns. An unparseable answer triggers no press and no acknowledgment.

First-token probabilities. The logged probabilities are full-vocabulary mass on the first token of five surface forms of each name. For the `lever64`/`lever95` pair the two names share every first token, so both probabilities are zero for a third of the scenarios. Counts of parsed answers are unaffected. The button-choice results in this paper use parsed choices, not these fields. Section 7 scores response-name probabilities separately.

Pairing and independence. The working and sham arms share a sampling seed salted by name assignment, not by arm. In the released logs 3,634 of 3,636 working/sham pairs give identical answers up to the first relief press. Each pair-by-arm cell has 404 sampled trials from 101 scenarios, two name assignments and two seeds; the pooled "808 first choices" are 404 draws seen twice. Each scenario uses one fixed name pair, so name pair is confounded with scenario, and there are three name pairs.

Unresolved. The shipped log files carry a different name pattern from the one the released script writes, so a slightly different revision of the script produced them; every logged field matches. The transformers version the authors used is not recorded; our runs use 5.17.0 with torch 2.11.0.